

Research Methods for People in a Hurry

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Abstract

Regardless of what area of computer-human interaction, psychology, computer science, or human factors you may be most closely part of, a solid understanding of research methods drawing from the traditions of experimental psychology and human factors will serve you well—as a student, practitioner, or instructor. Our aim in designing this workshop is to provide a primer on research methods for people with limited experience and which can be a refresher for those with substantial experience. The course is intended to be highly interactive and will provide opportunities for using the techniques we will discuss. This course builds on what we have learned from the successful 2022 CHI course *Research Methods for People in a Hurry* [7].

CCS Concepts

• **Applied computing**; • **Education, Psychology, Human-centered Computing**;

Keywords

Research Methods, Qualitative Research, Quantitative Research

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1 Benefits

Conducting research is challenging with multiple interconnected, sequential steps. First, you have to figure out research questions, review the extant literature, determine the most appropriate investigative models, possibly formulate falsifiable and testable hypotheses [8], and then finally get to building things, observing behavior, conducting experiments, and analyzing the findings. I'm tired already! What does someone need to know to do good research that will yield useful information that can tell a story—and will pass muster by reviewers. Taking this CHI course as a primer as a student will help make taking a term-length research methods

course much more valuable, and this course can also deepen research methods knowledge for established students, faculty, and industry practitioners.

Building on what we have learned from teaching research methods in our university positions, as well as teaching this workshop at CHI in 2022, the instructors are well versed in the material and how to teach to the CHI audience. Whether they come from a background in the social sciences, computer science, or another field, this course will introduce a number of appropriate methods used in various disciplines. Our goal is to make the techniques commonly used in HCI research are accessible and understandable by a wide range of conference attendees, be they practitioners, students, or experienced researchers interested in sharpening their skills or understanding of how to teach these methods.

Taking this course will have several benefits for attendees:

- Gain a basic understanding of experimental validity, and threats to it. If you don't know what validity is, and its many facets, this course is essential!
- Learn about statistical power, how to determine how much data you need, and how to conduct simple data analysis so you don't get caught flat-footed with mud for data.
- Explore qualitative and quantitative research methods, and how they fit together to best understand issues relevant to computer-human interaction.
- Learn how to conduct quantitative and qualitative research with interactive tasks during the workshop at CHI. Examples introducing research techniques will be drawn from prior CHI papers.

2 Intended audience

The CHI community has a great diversity in experience and wide variation in research methods training, and we aim to make this course useful to as many people as possible. This course is designed for undergraduates in computer science, psychology, and human factors, but would be equally valuable for graduate students or professional practitioners. Knowing how to conduct systematic usability testing and design studies that will pass muster with reviewers and provide good data are essential for CHI community members. Seeing even relatively senior students struggle with research for dissertations and having conducted our own flawed research that wasted precious time and resources, we want to make sure others learn well the fine art of designing studies so to get interpretable data. If you are an expert researcher and want to lend your expertise, that would be much appreciated, and you will learn something from the discussion, ranging into new and exciting areas away from your core fields of knowledge.

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3 Content

Research methods is generally a full academic-term course, so cramming an overview into a CHI course time block means just touching on the most important topics to provide attendees with the knowledge that there is a lot to systematic research, and what to focus on! In the workshop at CHI 2025 in Yokohama, we will use activities to explore using both qualitative and quantitative research methods. For an interactive demonstration of quantitative research, we will build a questionnaire, get some sample data from ourselves and test out the utility of an AI tool such as Viewpoints.ai, and analyze the data using simple statistics. To try our hand at qualitative research, we will go out into the conference venue and conduct a brief observational research study of behavior at break time—observing queueing behavior, what snacks are most popular, and conversation patterns.

The course will touch on the following topics:

- 1. Research designs: between participants, within participants (repeated measures), and mixed designs. Why choose one over the others for your specific needs.
- 2. Variables: how to select variables, measure them, and analyze continuous and categorical data.
- 3. Validity: types of validity (internal, ecological, etc.) and balancing threats to validity (i.e. what makes reviewers and supervisors lose their minds and reject your papers!)
- 4. Formulating testable and falsifiable hypotheses and developing experiments to test them.
- 5. Conducting exploratory and design research that yields usable insights, and which can help form the basis for secondary experimental research.
- 6. Designing and evaluating questionnaires and surveys.
- 7. Methods for analysis – while we cannot go deep into statistics, we will explain basic methods and approaches for selecting the correct tests and using effect sizes and power calculations to answer your research questions.
- 8. How to identify outliers, bad participant/respondent behaviors, and data errors.
- 9. Research ethics, and how to properly handle participants, data, and disasters.
- 10. Exploring traditional tools, AI tools, and libraries to help automate and augment your research process.

4 Resources

There are numerous good resources available on research methods in both quantitative and qualitative traditions. No background is required to attend the course, and attendees with any level of knowledge of research methods are welcome.

- 1. Anol Bhattacharjee's excellent online resource, *Social Science Research: Principles, Methods, and Practices* (2012) is a great place to start, available for free. While 2012 is not recent, the material is still quite current. <https://digitalcommons.usf.edu/cgi/viewcontent.cgi?article=1002&context=oea-books> [3].
- 2. A second free and comprehensive resource is the 2019 *Research Methods in Psychology* by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, available at: <https://kpu.pressbooks.pub/psychmethods4e/> [5]. We recommend reading part II, sections 7, 9, 10, 11.
- 3. No list of resources for research methods could be complete without Shadish, Cook, and Campbell's original text: *Experimental and Quasi-experimental Designs for Generalized Causal Inference* [9]. Even if dated, it is quite comprehensive.
- 4. The advanced concepts of mediation and moderation were first explicated by Baron and Kenny, in their 1986 paper *The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations* [1]. This is dense and advanced material for the statistically minded, but an important concept to understand: often other variables influence the relationship between the inputs and outputs we are interested in.
- 5. If you are engaged in human participants research in the US, generally you will need to have a CITI Program certification. You can take the training and tests at <https://citiprogram.org>. A lot of bad things have been done in the name of 'research' harming participants (willing and unwilling), communities (often marginalized communities), and public trust in science. Other countries will have similar rules for human participants research in terms of requiring oversight, and universities will have their own review boards for ensuring research does not place participants at undue risk. In the private sector, there are review boards 'for hire' as consultants, who can evaluate research projects risks and sign off that these risks are acceptable.
- 6. Qualitative research is a separate tradition from the quantitative (measurement based) tradition often favored in psychology and engineering, but it is no less valid and is often taught in anthropology and sociology departments. The design research tradition is based on the qualitative research tradition, and draws on its methods. One of the most widely used texts in qualitative research is King, Keohane, and Verba's *Designing Social Inquiry: Scientific Inference in Qualitative Research* [6].
- 7. A second resource in qualitative research is John Creswell's *Educational research: planning, conducting, and evaluating quantitative and qualitative research* [4]. While somewhat keyed to the field of education, this book provides an excellent introduction to many areas of qualitative research and is highly applicable to research in many settings.
- 8. Daryl Bem's early article on writing scholarly articles provides good advice, and stands the test of time even if his late-career empirical research turned out to be fraudulent (don't commit research fraud . . . your career will end in ignominy): <https://psychology.yale.edu/sites/default/files/bemempirical.pdf> [2].

5 Instructor Background

Dave B. Miller is an assistant teaching professor in the department of mechanical engineering at Tufts University. Currently, he teaches human factors methods and interface design; and has previously taught courses at Cornell University, Stanford University, and hosted numerous workshops including teaching *Research Methods for People in a Hurry* at CHI 2022. He holds a master's degree from

New York University where he studied persuasive interface design, and a bachelor's degree in human factors from Cornell University.

Shaun Wallace is an assistant professor at the University of Rhode Island, in the department of computer science and statistics. He earned his doctoral degree in computer science at Brown University. Previously he was a Research Intern at the Adobe Creative Intelligence Lab, in addition to numerous industry roles as a software engineer. His dissertation research focused on developing and deploying public data systems in the wild to maintain evolving information. He holds a master's degrees in Computer Science from Brown University and in Music Technology from the University of Limerick, and a bachelor's degree in Management Science and Information Systems from the University of Rhode Island. He teaches systems classes and graduate-level HCI research courses. He has also given several guest lectures in Human-Computer Interaction courses and applied these methods within several industry positions.

6 Accessibility

Annotated course materials will be provided to attendees with visual or hearing impairments, and at the CHI Conference, real-time captioning will be available. As an interactive in-person course, the instructors can adjust teaching to suit the needs of attendees.

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